Bode Wrap-Up

EGR 345

GVSU

Announcements

- end of semester schedule posted
- HW 5 is up
- quiz 2 this Thursday (11/12)
- exam 2 next Thursday (11/19)
    - exam 2 topics posted

Conceptual Review

- what is a Bode plot?
- where do Bode plots come from?
- what is the key assumption behind Bode plots/frequency response?

Frequency Response

- what do we mean by frequency response?
  - what do we mean by response?

Frequency Response

- what do we mean by frequency response?
  - what do we mean by response?

$$G(s) = \frac{Y(s)}{U(s)}$$

- $u(t) = ?$
- $y(t) = ?$

Sketching Review

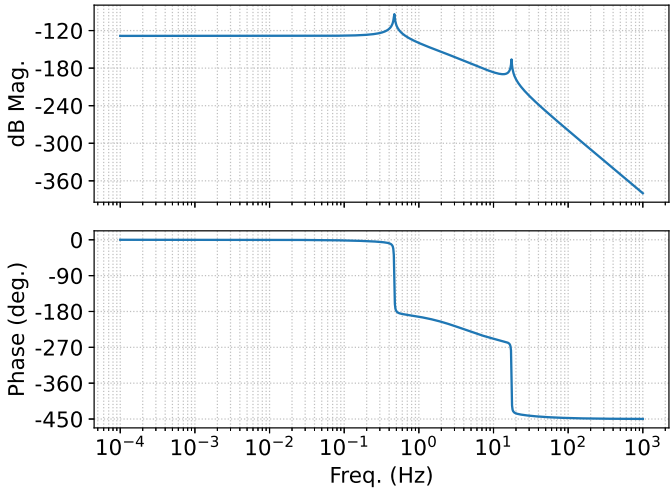
$$G_{sketch} = \frac{s + 29.33}{s(s + 0.08928)(s^2 + 1.186s + 1.989)(s + 407)}$$

Sketching Answer Check

```
num = [ 1. ,29.329]
den = [  1. ,408.308,521.248 ,853.055, 72.29, 0.]
G = control.TransferFunction(num,den)
```

System ID Review

- find the TF



System ID Answer Check

`bode_id_001_11_09_2026_07_49PM.csv`